# Freight Rate Prediction

Machine Learning Engineer Assessment | Technical report

A histogram gradient boosting model predicts total posted freight rates. On the September-October 2025 holdout, the selected model achieved MAE of $120.17 and WAPE of 5.02%, reducing MAE by 53.23% relative to a distance-only baseline. These are local holdout results; November-December target prices are not available.

## Data and quality checks

The labeled dataset contains 48,000 loads dated January 1 to October 31, 2025. The final prediction dataset contains 12,000 loads dated November 1 to December 31. The target is posted_rate in dollars. load_id is retained for output matching and excluded from the model.

| Issue | Labeled data | Final inputs |
|---|---|---|
| Missing weight | 300 | 165 |
| Non-positive weight | 292 | 145 |
| Missing market_index | 374 | 249 |
| Duplicate rows or load IDs | 0 | 0 |

Non-positive weights are replaced with missing values, resulting in 592 missing training weights and 310 missing final-input weights. The estimator handles missing numeric values natively. No loads are dropped, and high target prices are not removed solely because they are extreme. All parsed dates are valid; distances and labeled prices are positive. The labeled maximum price is $25,533, compared with a 99th percentile of $5,972.83.

## Chronological validation

| Stage | Training period and rows | Evaluation period and rows |
|---|---|---|
| Model selection | Jan-Jun: 28,806 | Jul-Aug: 9,671 |
| Final holdout | Jan-Aug: 38,477 | Sep-Oct: 9,523 |
| Final refit | Jan-Oct: 48,000 | Nov-Dec: 12,000 unlabeled |

Chronological splitting matches the task of predicting later loads from earlier observations. Encoders are fitted only on the training portion at each stage. MAE on July-August selects the primary model. The selected configurations are fixed before the September-October evaluation; holdout results do not trigger model changes. Final models are then refitted on all labeled rows.

## Features and model settings

The primary model uses pickup, delivery and equipment as categorical features; distance, weight, four coordinate columns and market_index as numeric inputs; and weekday, day of month, days since January 1, annual sine/cosine terms and inverse distance as derived inputs. This produces 16 features. OrdinalEncoder maps categories using training data, with unknown categories mapped to missing values. The estimator is explicitly told which columns are categorical, so city codes are not treated as ordered measurements.

HistGradientBoostingRegressor predicts total dollars directly using absolute_error loss, learning_rate=0.06, max_iter=250, max_leaf_nodes=23, min_samples_leaf=45, l2_regularization=2.0 and random_state=42. Early stopping is disabled to avoid an internal random validation split. Absolute-error loss prioritizes typical dollar error and reduces sensitivity to extreme targets; it does not remove extreme observations.

## Model selection results

| Jul-Aug candidate | MAE $ | RMSE $ | WAPE % |
|---|---|---|---|
| Distance baseline | 249.01 | 674.72 | 10.47 |
| Without quote_signal | 137.34 | 623.22 | 5.78 |
| With quote_signal | 139.65 | 626.29 | 5.87 |
| December-compatible model | 171.15 | 629.11 | 7.20 |

Adding quote_signal lowers July MAE from $132.72 to $127.39 but increases August MAE from $142.12 to $152.31. Its combined MAE is worse, so the selected primary model excludes it. This comparison supports the decision for this experiment; it does not establish that the feature is always unhelpful.

## Final holdout results

| Sep-Oct model | MAE $ | RMSE $ | WAPE % |
|---|---|---|---|
| Distance baseline | 256.95 | 684.25 | 10.74 |
| Selected main model | 120.17 | 638.06 | 5.02 |
| December-compatible model | 111.63 | 634.66 | 4.67 |

The December-compatible model performs better on this holdout, but the primary model remains the July-August winner. This variation is reported rather than using the holdout to reselect the model. All metrics include every holdout load, including large-error cases.

MAE is mean absolute dollar error. RMSE is the square root of mean squared dollar error and is more sensitive to large misses. WAPE is 100 times total absolute error divided by total absolute actual price; it is not an accuracy percentage. The gap between MAE and RMSE indicates that large errors remain.

## December scenario

The supplied scenario fixes Lexington to Fort Wayne, 360 miles, Dry Van and 32,000 lb; only the date changes across December 2025. Coordinates, market_index and quote_signal are unavailable. A separately trained model uses the 11 available and derived features, with the same estimator settings. It is evaluated before final refitting. No unavailable market signals are invented.

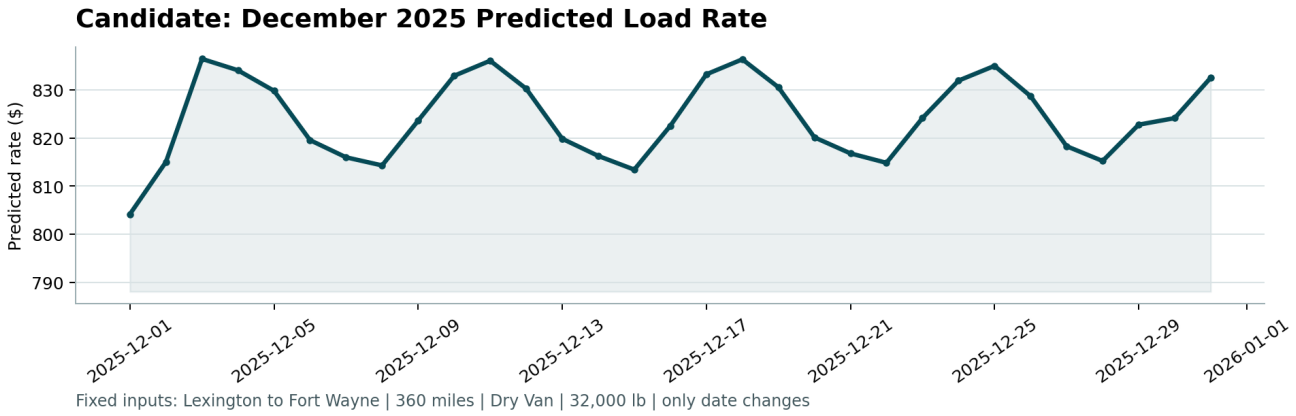



Figure 1. Original chart produced by the supplied score.py using the submitted December predictions. Predicted rates range from $804.17 to $836.41. The recurring pattern reflects the learned date effects, not observed December prices or a guaranteed future cycle.

## Reproducibility and output validation

The repository includes a cleaned notebook and matching run.py. Run python -m pip install -r requirements.txt, place the original CSV inputs in data/, then run python run.py. The workflow trains, evaluates, saves both prediction files, and calls the original score.py. Predictions are matched by load_id and written with two decimal places. The scorer accepts exactly 12,000 unique required IDs with positive finite predictions, and checks all 31 fixed December rows.

The original Colab run used NumPy 2.1.3, pandas 2.2.3, Matplotlib 3.10.0, scikit-learn 1.6.1 and threadpoolctl 3.7.0. Exact recorded versions are included in results/environment_versions.txt. The supplied scorer validates outputs and creates the chart; employer-side hidden-set metrics are calculated after submission.

## Limitations and further work

Only one year is partially observed, so annual seasonality and holiday effects are not established. Tree models have limited ability to extrapolate beyond the training date range. The fixed December lane has no labeled December outcomes for direct validation, and its reduced feature model omits observed market conditions. Coordinate semantics and signal provenance were not independently verified. Further work would use additional historical years, rolling-origin evaluation and verified feature definitions; the final holdout would remain separate from tuning.

Sources: supplied assessment instructions and scorer; original input CSVs; executed Colab notebook; exported model-selection and holdout metrics, row-level holdout predictions, final predictions and scorer chart.